Low-Temp White 130 °C

Low-temperature cure polyester, 130 °C · RAL 9010

RAL 9010 · Pure white

Product	Code • series	Colour reference	Document
Low-Temp White 130 °C 130 °C 低温白	BL-L301 BL-L Low-Temperature Cure	RAL 9010 · Pure white	TDS BL-L301 Rev. 2026-10

P1000 (low-cure) · test items on request — no product certificate held

1. Description

Cures at 130 °C so MDF edges do not char and assemblies with plastic inserts survive the oven. White is the hardest low-cure colour: it yellows with overbake, so the oven profile matters more than the powder. Metal parts are possible, but the lower cure temperature leaves less margin.

2. Typical properties

Property	Value	Method
Chemistry	Low-temperature cure polyester, 130 °C	—
Colour reference	RAL 9010 Pure white — approval on a sprayed panel against a physical chip	RAL
Finish and gloss	30–80 @60°	ASTM D523
Film build	60–80 µm	ISO 2808
Density	1.45–1.55 g/cm³	ISO 8130-2/3
Coverage	9.5 m²/kg at 100 % film · 9.0 m²/kg planning value at 95 % utilisation (with reclaim)	calculated
Cure schedule	130 °C × 20 min (object temperature)	object temp., not oven set point
Particle size D50	30–45 µm	ISO 8130-1 / -13
Stoving loss	≤ 1–2 %	ISO 8130-7
Storage stability	1 month at 35–40 °C	ISO 8130-8

3. Performance data

Test	Method	Typical result
1. Adhesion, cross-cut	ISO 2409 / ASTM D3359	0–1
2. Impact, direct and reverse	ASTM D2794	50 kg·cm on metal; on MDF, film integrity per trial
3. Pencil hardness	ASTM D3363	H
4. Gloss 60°	ASTM D523	30–80 per specification
5. Colour difference	ASTM D2244	ΔE ≤ 1.5 same batch
6. Neutral salt spray	ASTM B117 / ISO 9227	500 h (metal substrates)
7. QUV-A	ASTM G154	500 h
8. Surface hardness on MDF	EN 12720 (test item)	On request — depends on board and film build

Measured to the method listed, on production batches or the third-party report held for this system. Not a certificate: acceptance values are fixed in the trial report and the purchase specification.

4. Application and cure

01 MDF preparation

Sand 120–180 grit, moisture 6–8 %, edges profiled and dust-free. Preheat the board to 40–50 °C before spraying so the powder adheres and the edges hold film.

02 Metal preparation

Degrease → conversion coating or phosphate → DI rinse → dry, as for the standard-cure products.

03 Application

Corona 40–70 kV on MDF (lower voltage limits back-ionisation on a non-conductive board), 100–200 mm, 70–90 µm.

04 Cure

Object temperature 130 °C for 20 min. Measure the board core, not the surface: the face can reach 130 °C while the core is still at 110 °C.

05 Oven control

Uniformity within ±5 °C. Ten degrees over for ten minutes is enough to shift a white, and there is no polishing that back.

5. Packaging, storage and ordering

1. Standard pack: 20 kg net, polyethylene-lined carton. Other pack sizes on request.
2. Store below 30 °C, dry, out of direct sunlight, in the unopened pack. Keep away from ovens and hot lines.
3. Shelf life: 12 months from the production date in the original pack. Storage stability is verified at 35–40 °C for one month (ISO 8130-8).
4. A batch test report accompanies every delivery; a retained sample is kept for 12 months.
5. Sample in 3 days; production lead time confirmed per colour and pack size at order.
6. Colour match ΔE ≤ 1.5 within a batch. Cross-batch tolerance is agreed in the purchase specification.
7. Ask for: SDS · third-party test report · batch test report · physical colour chip.

Low-cure note: this chemistry is more storage-sensitive than standard powder — keep packs sealed and cool, and re-check gel time after a month open.

6. Health and safety

1. Not classified as dangerous goods for transport. The SDS is available for every product — ask for it with the TDS.
2. The powder is a combustible dust. Earth the workpiece, the gun and the booth; no open flame or hot work in the booth; extract booth air; avoid dust clouds when emptying or cleaning.
3. Wear a P2 particulate mask, goggles and gloves when handling dry powder or cleaning the booth.
4. Cure volatiles: extract the oven and let parts cool before handling. Do not enter the oven zone without ventilation.

7. Also in this range

Code	Product	Colour	Film build	Cure
BL-M102	Bonded Gold Sand	RAL 1036	70–90 µm	200 °C × 10 min
BL-M103	Bonded Mirror Silver	RAL 9006	60–80 µm + clear	180–200 °C × 10–15 min
BL-A201	Outdoor Sand Grey	RAL 7035	60–80 µm	200 °C × 10 min

8. Document control

Sheet	Revision	Issued	Supersedes	Languages
BL-L301 TDS	2026-10	2026-10	None — first issue	English + 中文 (4 pp.)

9. Limits of this document

What this product is not for:

— A 130 °C film has less cure margin than a 200 °C film. If the part later sees 160–180 °C in service, tell us — post-cure can crack the film.

— Low-cure powder is more storage-sensitive: keep cartons sealed and cool, and check gel time if a pack has stood open for a month.

— Approve white on a sprayed MDF panel, not on a metal chip — the same powder reads differently on board.

1. Values in this sheet are typical for the product and were measured to the methods listed. They are not a certificate and not a warranty of fitness for a particular purpose. Acceptance values are fixed in the trial report and the purchase specification.
2. Colour: RAL references only. RAL does not define sRGB values, so the swatch in this sheet is a digital approximation — approval is made on a sprayed panel against a physical RAL chip.
3. No AAMA 2603 / 2604 / 2605, QUALICOAT, GSB or UL product certificate is held. We test to those methods and can supply third-party reports and in-house batch reports; if a tender requires the certificate itself, we will say plainly that we do not have it.
4. The 4,000 h salt spray and 10-year South Florida results shown elsewhere on our site belong to the fluorocarbon (PVDF) system, not to this powder product.

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Ask for: SDS · third-party test report · batch report · physical colour chip · sample in 3 days

130 °C 低温白

低温固化聚酯，130 °C · RAL 9010

RAL 9010 · 纯白

产品	编号 • 系列	参照颜色	文件
Low-Temp White 130 °C 130 °C 低温白	BL-L301 BL-L 低温固化	RAL 9010 · 纯白	TDS BL-L301 Rev. 2026-10

P1000（低温型）· 可按测试项目送检（可索取报告）——不持有产品认证证书

1. 产品描述

130 °C 固化，MDF 不焦边，含塑料嵌件的组件也能过炉。白色是最难的低温色：过烘就黄，所以烘道曲线比粉末本身更关键。金属件也能做，但低温固化留下的工艺余量更小。

2. 典型性能

项目	数值	方法
体系	低温固化聚酯，130 °C	—
参照颜色	RAL 9010 纯白——以喷板对照实物色卡确认	RAL
表面与光泽	30–80（60°）	ASTM D523
膜厚	60–80 µm	ISO 2808
比重	1.45–1.55 g/cm³	ISO 8130-2/3
覆盖率	100% 成膜 9.5 m²/kg · 按 95% 利用率（含回收）规划值 9.0 m²/kg	计算值
固化条件	130 °C × 20 min（工件温度）	测工件温度，非炉膛设定
粒径 D50	30–45 µm	ISO 8130-1 / -13
烘烤失重	≤ 1–2 %	ISO 8130-7
贮存稳定性	35–40 °C 下 1 个月	ISO 8130-8

3. 性能数据

检测项目	方法	典型结果
1. 划格附着力	ISO 2409 / ASTM D3359	0–1
2. 正反冲击	ASTM D2794	金属件 50 kg·cm；MDF 按试板测涂层完整性
3. 铅笔硬度	ASTM D3363	H
4. 光泽 60°	ASTM D523	按规格 30–80
5. 色差	ASTM D2244	同批 ΔE ≤ 1.5
6. 中性盐雾	ASTM B117 / ISO 9227	500 h（金属基材）
7. QUV-A 加速老化	ASTM G154	500 h
8. MDF 表面硬度	EN 12720 (test item)	可送检——取决于板材与膜厚

按所列方法在量产批次或本体系所持第三方报告上测定。不是证书：验收值以试粉报告与采购规格为准。

4. 施工与固化

01 MDF 前处理

砂纸 120–180 目，含水率 6–8%，边角成型并除尘。喷前将板面预热至 40–50 °C，粉末才挂得住、边角才上膜。

02 金属件前处理

脱脂 → 转化膜或磷化 → 纯水洗 → 干燥，与标准固化产品相同。

03 喷涂

MDF 上电晕 40–70 kV（板材不导电，电压低一点可减少反向电离），枪距 100–200 mm，膜厚 70–90 µm。

04 固化

工件温度 130 °C × 20 min。测板材芯部温度，不要测表面：表面到 130 °C 时芯部可能只有 110 °C。

05 烘道控制

温度均匀性 ±5 °C 以内。超温 10 °C 十分钟，白度就变了，抛不回来。

5. 包装、贮存与订货

1. 标准包装：20 kg 净重，纸箱内衬聚乙烯袋。其他包装规格可订。
2. 30 °C 以下、干燥、避免阳光直射，保持原包装未开封。远离烘道与高温工位。
3. 保质期：原包装自生产日起 12 个月。贮存稳定性按 ISO 8130-8 于 35–40 °C 验证一个月。
4. 每批附批次检测报告；留样保存 12 个月。
5. 3 天出样；生产交期按颜色与包装规格在订单时确认。
6. 同批色差 ΔE ≤ 1.5；跨批公差在采购规格中约定。
7. 可索取：SDS · 第三方检测报告 · 批次检测报告 · 实物色卡。

低温型提示：该体系比标准粉更怕贮存——保持密封阴凉；开袋超过一个月要复测胶化时间。

6. 健康与安全

1. 不属于运输危险品。每个产品均提供 SDS（安全数据表），可与 TDS 一并索取。
2. 粉末为可燃性粉尘。工件、喷枪与喷房必须接地；喷房内禁止明火与动火作业；喷房须抽风；倒粉与清房时避免扬尘。
3. 处理干粉或清理喷房时，佩戴 P2 防尘口罩、护目镜与手套。
4. 固化挥发物：烘道须抽风；工件冷却后再搬运。无通风不得进入烘道区域。

7. 同系列其他产品

编号	产品	颜色	膜厚	固化
BL-M102	邦定金砂	RAL 1036	70–90 µm	200 °C × 10 min
BL-M103	邦定镜面银	RAL 9006	60–80 µm + clear	180–200 °C × 10–15 min
BL-A201	户外砂纹灰	RAL 7035	60–80 µm	200 °C × 10 min

8. 文件控制

文件	版本	签发	替代版本	语言
BL-L301 TDS	2026-10	2026-10	无——首版	英文 + 中文（共 4 页）

9. 本文件的边界

这款产品不适合用在：

— 130 °C 涂层的固化余量小于 200 °C 涂层。工件在服役中会到 160–180 °C 时必须告知——后期固化会让涂层开裂。

— 低温粉对贮存更敏感：纸箱密封、阴凉存放；开袋放置一个月以上要复测胶化时间。

— 白色要在喷好的 MDF 板上确认，不要用金属试片——同一款粉在板材上的观感不同。

1. 本表数值为该产品的典型值，按所列方法测定。它们不是证书，也不构成对特定用途适用性的保证。验收值以试粉报告与采购规格为准。
2. 颜色：仅作 RAL 参照。RAL 未定义 sRGB 数值，本表色块为数字近似——以喷涂样板对照实物 RAL 色卡确认。
3. 我们不持有 AAMA 2603 / 2604 / 2605、QUALICOAT、GSB 或 UL 产品认证证书。我们按这些方法检测，可提供第三方报告与厂内批次报告；若招标硬性要求证书本身，我们会直接说明没有。
4. 我们网站上 4000 小时盐雾与 10 年南佛罗里达曝晒的结果属于氟碳（PVDF）体系，不属于本粉末产品。

签发：广东宝利莱工业有限公司 · jerry888.xu@gmail.com · +86 188 2310 0890
可索取：SDS · 第三方检测报告 · 批次报告 · 实物色卡 · 3 天出样